Perceptron Numerical Guide

Ye guide ANN aur Perceptron ke numericals ko solve karne ke liye banayi gayi hai. Isme aap seekhenge:
• Perceptron kya hota hai
• Error kaise nikalte hain
• Learning rate (a) kya hoti hai
• Weight update kaise hota hai
• OR function numerical step-by-step
• XOR kyun solve nahi hota

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1. Perceptron Basics

Perceptron ANN ka basic neuron hota hai jo binary classification karta hai. Iska output normally 0/1 ya -1/+1 hota hai.

Main Formula:
y = Step(w1x1 + w2x2 + b)

2. Important Terms

Symbol	Meaning
x1, x2	Inputs
w1, w2	Weights
b	Bias
a	Learning Rate
e	Error
y	Output

3. Error Kaise Nikalte Hain?

Error nikalne ka formula:

e = Expected Output - Actual Output

Example:
Expected = 1
Actual = 0
Error = 1 - 0 = 1

4. Learning Rate (a)

Learning rate decide karti hai ke weights kitni speed se update honge.

Choti learning rate = slow learning
Bari learning rate = fast learning

5. Weight Update Formula

wi = wi + (a × xi × e)

Jahan:
wi = old weight
a = learning rate
xi = input
e = error

6. Complete Numerical Example (OR Function)

x1	x2	Output
0	0	0
0	1	1
1	0	1
1	1	1

Initial Values:
w1 = -0.2
w2 = 0.4
a = 0.2

Step 1:
x1 = 0, x2 = 0
y = (-0.2×0) + (0.4×0) = 0
Output correct → No update

Step 2:
x1 = 0, x2 = 1
y = (-0.2×0) + (0.4×1) = 0.4
Output = 1 → Correct

Step 3:
x1 = 1, x2 = 0
y = (-0.2×1) + (0.4×0) = -0.2
Output = 0 (Wrong)

Error:
e = 1 - 0 = 1

Weight Update:
w1 = -0.2 + (0.2×1×1) = 0
w2 = 0.4 + (0.2×0×1) = 0.4

New weights:
w1 = 0
w2 = 0.4

7. XOR Problem

Single perceptron XOR solve nahi kar sakta kyunki XOR non-linearly separable hota hai. XOR solve karne ke liye Multi Layer Perceptron (MLP) use hota hai.

8. Exam Tips

Numerical solve karte waqt:
• Sabse pehle formula likho
• Inputs aur weights clearly likho
• Expected aur actual output compare karo
• Error calculate karo
• Weight update formula apply karo
• Final weights zaroor mention karo